

An Agricultural Machine Learning Framework for Ensemble Regressor-Based Crop Price Prediction

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Abstract— Indian economy is agrarian-based and employs more than half of the working people too, and indirectly contributes significantly to national GDP. Industry is, however, highly volatile to such fluctuations like crop price, which is a great economic risk for the producers but, on the other hand, makes markets highly inefficient. Traditional statistical analyses are also bound to miss the delicacy of nature and price movement driver interdependence. This article eschews such constraints by employing cutting-edge machine learning models, i.e., Decision Tree Regression and boosted Stacking Regressor architecture, in a manner that they are capable of predicting crop prices with success. The algorithm considers historical crop prices, weather conditions, quality of soil, and past market demand history records from farm and government databases. The proposed Stacking Regressor makes use of XGBoost, Random Forest Regressor, and Ridge Regression as the base learners and XGBoost as the meta-learner. The ensemble model can attain a high and consistent R^2 metric value of 91.36% and Mean Absolute Error (MAE) of just 0.18, which is very strong and can be practically applied effectively. This research finding brings a paradigm shift to agricultural economics in the backdrop that policymakers, farmers, and market specialists can make sound decisions with information. Future development of this project can also include real-time streams of data, deep learning models, and time-series forecasting into model accuracy and availability for various climates of agriculture.

Keywords—Decision Tree, feature engineering, Stacking Regression, Random Forest Regressor, XGBoost.

I. INTRODUCTION

Agriculture has remained a significant component of the Indian economic system and the primary means of livelihood for the majority of the population. Agriculture is a significant sector in the economy of the country with over 58% of the workforce in India involved in agriculture. But the fortunes of farmers lie in the vagaries of farm products, which vary not just in the Indian economy but globally [1]. In the last year, high food prices combined with increasing energy prices have drained consumers' purchasing power across the globe. The International Monetary Fund points

out that the share of their consumer expenditure spent on food items is greater in low-income countries than it is in high-income nations. In these developing nations, spending on food can reach as high as 44% of disposable income, while in emerging nations it averages at 28% and falls to 16% in developed economies [2]. While little research has been done on price forecasting for agricultural products, particularly vegetables, machine learning techniques are increasingly being applied to address forecasting problems in a wide range of applications [3].

The country's economic decision-makers must thus rely on a true crop production forecast in the process of acquiring useful exports and imports suggestions for the improvement of the food security of the country. Generally, many factors affect crop production, such as terrain, grade of the soil, pest issues, genotype, performance and availability of water, weather, harvesting method, and other elements [4]. An effective crop price forecasting model can provide opportunities to farmers that can benefit society in a greater sense [5]. The classical time sequence forecasting approach was found to have a high degree of accuracy in forecasting the time series data with linear properties. But as the market became increasingly complex, nonlinear and irregular step by step, this kind of methods resulted in tremendous trouble in the adjustment of the forecasting model's parameter and showed certain shortcomings in the prediction of the price. With the advent of artificial intelligence, the smart forecasting techniques like neural network, support vector machine and extreme learning machine had become the center of price prediction studies gradually.[6] Taking ML/DL based models into account, with a daily arrival of a new price data, refreshing the models could cause stability due to quality problems related to the crop price data.[7] The objective of this paper is to discover the availability of the models or frameworks to forecast and suggest the price for farmer's produce.[8] ARIMA, Regression Analysis, and Expert Systems are some of the old econometric and statistical packages which do not have big data handling capability and interdependence of drivers in their attempt to forecast prices. We thus employ Decision Tree Regression, which is

an ML method of constructing a model capable of properly estimating crop prices.

Unlike previous approaches, which either applied linear models on linear datasets or struggled when applied to nonlinear datasets, our work systematically explores both cases. We first implemented linear models on linear datasets, then evaluated their performance on nonlinear datasets to highlight their limitations. Finally, we combined linear and nonlinear approaches into a stacking ensemble framework. This hybrid strategy significantly improved predictive performance, achieving an accuracy of 91.36% (R^2) with $MAE = 0.18$, thereby demonstrating a clear advancement over traditional forecasting methods. The unique contribution of this study lies in formulating a hybrid machine learning framework for crop price prediction that integrates both linear and nonlinear approaches within a stacking ensemble.

XGBoost, although not performing optimally as an individual model, serves effectively as a meta-learner in the stacking ensemble, where it combines complementary patterns from other models to deliver high predictive accuracy. For this research, we utilized a Stacking Regressor, with XGBoost, Random Forest, and Ridge Regression as the base learners and XGBoost as the meta-learner. This design choice highlights how our model goes beyond traditional econometric and single-model machine learning methods by exploiting ensemble diversity.

The remainder of the paper is structured as follows. Section II describes the literature review conducted in this research. Section III provides a proposed methodology. Results and discussion is highlighted in section IV. Section V presents the conclusion and also identifies areas for future investigation.

II. LITERATURE REVIEW

The following is an analysis of the efforts taken by several authors to come up with methods to predict crop price. Sandhu Dutt et al. [1] examined AI methodologies for predicting farm prices. They highlighted the constraints of conventional methods such as ARIMA and exponential smoothing that don't work in the event that price action is nonlinear. The research cited a few of the AI methods, ranging from machine learning methods such as linear regression, random forests, to deep learning methods such as LSTM. The authors highlighted the potential of AI in making markets more transparent, maximizing resource utilization, and minimizing the risk for farmers and traders. The authors' study suggests the capability of AI in enhancing decision-making for agricultural actors while resolving ethical issues and policy challenges. Humaira Rana et al. [2] examined AI-driven methods for the prediction of crop price forecasts in Pakistan. Analysis replicated historical time series of prices (2007–2022) with the help of machine learning methods such as ARIMA, Random Forest, and Long Short-Term Memory (LSTM). Apache Spark was applied to pre-process the big data utilized upon analysis. The top-ranked model among the criteria of accuracy had a ranking value of $R^2 = 0.8$. The article cites the possibility of using AI to enhance decision-making by farmers, traders, and policymakers. Research in the future advises the use of weather data and sentiment

analysis of markets for enhanced price prediction. Chandan Sharma et al. [3] have given a machine learning model for the prediction of agricultural prices. The article focuses on the effect of climatic variables—temperature, humidity, pH, and rainfall—on crop yields and market prices. The authors applied Decision Tree Regression to predict prices based on past data as well as weather conditions. The proposed model provides time-series analysis for future price prediction. The study is a pointer to the future of AI in helping farmers with crop selection and budgeting, with optimization yet to arrive with more environmental and market factors for even more accuracy. G. Thapaswini and M. Gunasekaran et al. [4] discussed a survey on AI-based approaches towards agricultural price forecasting. The study establishes that traditional statistical models and that the models do not actually make predictions on crop prices are referenced. The writers promote the usage of machine learning methods such as decision trees and neuro-evolutionary methods to improve price predictions. The study work of the writers' centers around the application of AI to help improve agricultural output, improve strategy in the markets, and providing improved price proposals to farmers. The work also contains preprocessing methods for data and optimizations by genetic algorithms for standalone price prediction systems. Ranjani Dhanapal et al. [5] experimented with the integration of AI methodologies for the prediction of crop prices. The farmers are vulnerable to market price fluctuation and recommend the utilization of machine learning-based models for the prediction of future prices. The authors have used a Decision Tree Regressor to establish models for historic rainfall and Wholesale Price Index (WPI) data and accordingly predicted prices very accurately. It shows the efficiency of AI-powered predictive models to reduce farmers' economic losses and better market planning. Future upgrading involves the adoption of climate-resilient cultivation practices and fertilizer application. Guang Zheng et al. [6] constructed a wavelet transform and BP neural network-prediction model. The research was aimed at improving the accuracy of price prediction for farm produce through low- and high-frequency price trend decompositions. The authors utilized 72 months of vegetable prices and demonstrated that the hybrid model outperformed basic models like ARIMA. They have outcomes with the lowest mean absolute error of 0.083 yuan/kg, which validates its use in market planning. In addition to supporting further research, real-time data feeds are recommended to enhance dynamic price forecasting. Ayush Jain et al. [7] pointed out challenges like human errors in manually recorded crop prices and market fluctuations that affect forecast stability. To overcome this, they proposed examining data quality prior to using predictive models. Their research focused on identifying and correcting inconsistencies in order to enhance accuracy. Through the incorporation of data validation mechanisms, they sought to make forecasting more reliable, which would be useful for supply chain planners and government agencies to make better decisions for effective agricultural market management. Girish Hegde, Vishwanath R Hulipalled and J.B Simhaet al. [8] investigates AI-based methods to predict prices of agricultural commodities. The article refers to the traditional models like ARIMA and exponential smoothing, observing that they are deficient in handling nonlinear price behavior. The article highlights modern AI-based methods including machine learning models including decision trees, support vector machines (SVM), and deep learning models including LSTM. The writers propose AI as a way to enhance market efficiency,

reduce risks, and assist farmers and traders in their decision-making. The article also discusses such issues as data quality and ethical concerns in AI-based forecasting

models. Table 1 provides an overview of earlier research on crop price prediction, including its methodologies, applications, accuracy, and prospects for the future.

Table 1: Comparison of Existing System

Author	Year	Methodology/Framework	Application	Limitation/Future Scope
Sandhu Dutt et al. [1]	2024	AI-driven models like Linear Regression, Random Forest, LSTM	Forecasting agricultural prices of crops like TOP (Tomato, Onion, Potato) in India	Data scarcity and ethical concerns. Future work: Integrate satellite data, IoT, and improve transparency.
Humaira Rana et al. [2]	2024	Big Data Framework (Apache Spark, Pyspark), ML models (ARIMA, Random Forest, LSTM)	Predicting spinach prices in Pakistan using 17M records	LSTM outperformed; suggests future integration of big data and advanced deep learning models for other crops
Chandan Sharma et al. [3]	2023	Decision Tree Regressor, KNN, Random Forest, Neural Network, Data Mining, Time Series Analysis. Accuracy-91.70%	Price prediction of fruits, vegetables, and pulses based on weather parameters	Model trained on specific weather and crop data. Future scope: real-time deployment and wider crop variety prediction.
G. Thapaswini, M. Gunasekaran[4]	2022	Decision Tree, Neuro-Evolutionary Algorithms Accuracy-86%	Crop price prediction and productivity improvement	Future scope: Genetic Algorithm-based auto price recommendation
Ranjani Dhanapal et al [5].	2021	Supervised Machine Learning - Decision Tree Regressor Dataset: Rainfall and Wholesale Price Index (WPI) data from <i>data.gov.in</i>	Predict crop prices for the next 12 months to help farmers make better decisions and reduce losses	- Limited to past rainfall and WPI data only - Focused on 20 crops - Future scope: Add fertilizer suggestions, pest/disease warnings, integrate advanced AI models, and climate-aware farming techniques for better prediction accuracy
Guang Zheng et al. [6]	2020	Wavelet Transform + BP Neural Network combined model	Short-term price prediction of spinach, cabbage, tomato, pepper, and potato	Limited to 5 vegetables. Future scope: expand system for various crops and integrate more AI models.
Ayush Jain et al. (IBM Research) [7]	2020	Context-based Model Selection, LSTM, MLP, ARIMA, SARIMA, Data Quality Features	Crop price forecasting of Tomato and Maize across 14 markets in India	Data quality issues addressed. Future work: generalize model for other crops, improve robustness, and handle extreme
Girish Hedge, Vishwasnath R Hulipalled, J, B Simha [8]	2020	Machine Learning Models: Random Forest, Decision Tree, XGBoost, Clustering, Neural Nets	Forecasting and price prediction of agriculture commodities	Focus on dataset improvement and real-time market integration. Future scope: Build a full prediction system

III. METHODOLOGY

This research utilizes a Stacking Regression and XGBoost machine learning model of forecasting to predict agricultural crop prices. The research process includes systematic data pre-processing, training, and validation. A Stacking Regressor is utilized to model the various base models, i.e., XGBoost, Random Forest Regressor, and Ridge Regression, and then to add predictive strength. XGBoost comes in handy as it handles missing values,

accommodates high-dimensional data, and discerns sophisticated patterns in crop price fluctuations. With inputs of historical trends, weather patterns, and demand from the market that have been inputted to the model, so firm and sturdy conclusions can be made so farmers and policymakers can make informed choices

This figure 1 shows the suggested system's workflow. It demonstrates the entire procedure, from gathering and preprocessing data to training models and forecasting crop prices in the end.

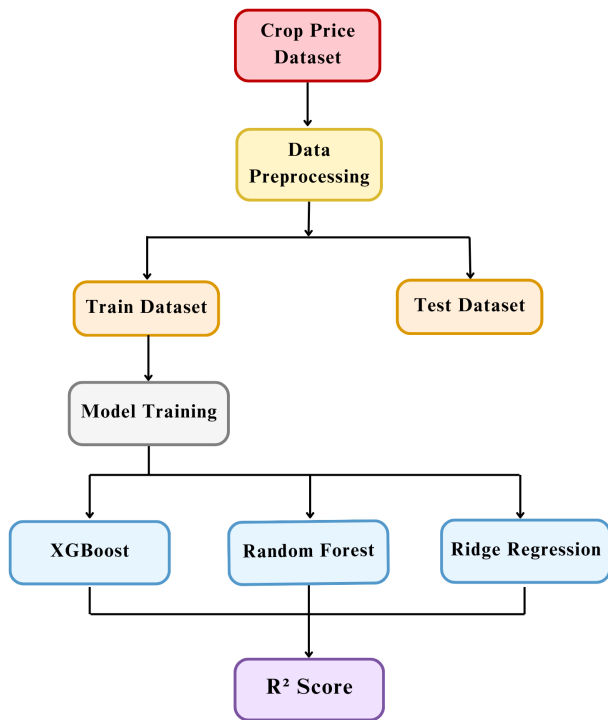


Fig1. Proposed Methodology

1) Data Collection:

Data were split into two sets: 80% for training the model to identify concealed patterns and 20% for testing in order to confirm the model's performance on new data. We employed the “Crop Prediction” dataset on Kaggle [17] for this study. The dataset includes soil nutrient information (N, P, K), pH levels, weather conditions, and crop information. It is lightweight, approximately 199 KB in size, which makes it simple to manage and preprocess during training and testing of models.

The most important variables are:

1. Historical Crop Prices – Past market prices of different crops.
2. Soil Fertilizer Concentrations – Levels of Nitrogen (N), Phosphorus (P), and Potassium (K) in soil.
Weather Conditions – Temperature, humidity, and rain records.
3. Soil pH Levels – Soil alkalinity or acidity readings, which influence crop production.
4. Market Demand and Supply – Demand and supply of crops between different.

2) Data Pre-processing:

Prior to training the model, the data is preprocessed as needed. The missing values are imputed with mean/mode or removed if needed. Feature engineering is performed by using polynomial transformation and log transformation to achieve a good feature representation. The numerical features are standardized using Standard Scaler so that all variables contribute equally. Removal of outliers is carried out by Winsorization in order not to allow extreme values to bias predictions. Lastly, feature elimination is carried out through Variance Inflation Factor (VIF) and correlation analysis for the elimination of redundant or irrelevant features, as indicated in Fig.2. It lists the main characteristics that are used, such as soil nutrients, pH, weather, past prices, and market demand. These steps

ensure that inconsistencies in the dataset do not negatively impact model performance, leading to a more accurate and reliable predictive model.

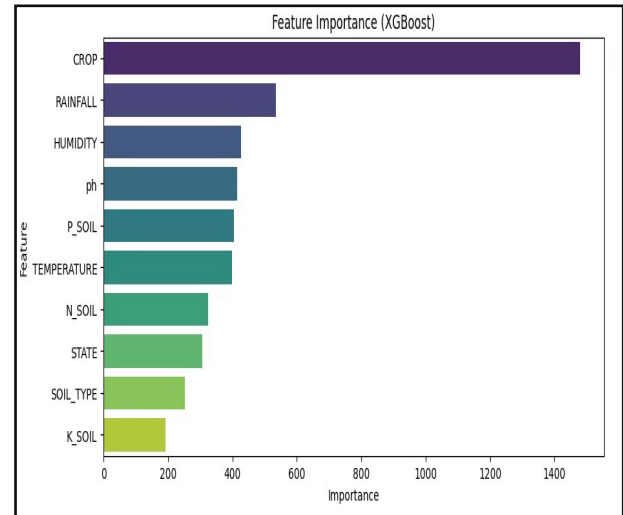


Fig. 2. Feature Description

3) Data Modelling:

The following base models are used to leverage each of their strengths and enhance overall performance:

A. Random Forest Regressor

- A bagging-based ensemble algorithm that reduces variance and improves robustness
- By averaging a large number of decision trees, Random Forest Regressor can detect complicated interactions between features and targets
- Its ensemble framework prevents overfitting and generates a more precise prediction

B. Ridge Regression

- A non-overfitting linear model using L2 regularization.
- Ridge Regression introduces a penalty to the loss function so that it no longer has gargantuan weights and leads to a reduced model.
- Because it is linear, it is particularly best for very linearly correlated data. The outcome of base models is passed through a meta learner in order to obtain the final price forecasts.

C. XGBoost Regressor

- Trained on top of base models' output so that they understand the strengths and weaknesses of a single model.
- By consensus of prediction of multiple models, the XGBoost Regressor meta learner makes a more refined and accurate final prediction.
- Its ability to learn from the output of the base models leads it to identify complex relationships and make better-informed predictions.

4) Model Evaluation

To measure the model's performance, the following measures are utilized:

- Mean Absolute Error (MAE): Calculates the average absolute amount of prediction error, i.e., how much predicted prices $\hat{y}_i$ differ from actual values y_i , as defined in equation (1)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

- R²Score (Coefficient of Determination): Estimates how much variance in actual prices is explained by the model, with smaller values closer to 1 indicating better accuracy, as given in equation (2).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

- Residual Analysis – Compares the actual and predicted values to determine the difference and identify model biases.

IV. RESULTS AND DISCUSSION

Figure 3 presents a comparative analysis of four regression models—XGBoost, Ridge Regression, Random Forest Regressor, and a Stacking Model—based on their **R² Scores**. The R² score reflects the proportion of variance in the crop price data that the model can explain. A higher R² indicates better model performance. The results are summarized in Table 2.

Table 2: Accuracy of different Model

Model	R ² Score	MAE
XGBoost	30.76%	0.32
Ridge Regression	46.79%	0.26
Random Forest Regressor	76.12%	0.21
Stacking Model	91.36%	0.18

From the findings, one can see that the Stacking Model performs better than the single models, creeping up to the highest R² Score of 91.36%. This shows that the model is able to account for a great deal of variability in crop prices and therefore is highly appropriately suited for use in predictive applications in agricultural markets. Additionally, the Mean Absolute Error (MAE) for the stacking model is only 0.18, further attesting its validity. Lower MA values that indicate smaller differences between predicted and actual values, showing the robustness and

stability of the model.

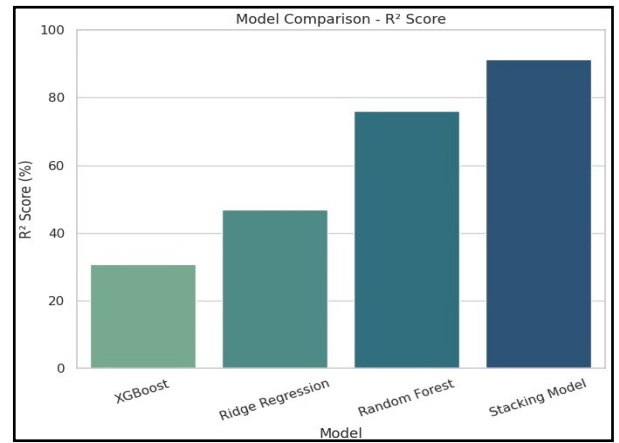


Fig.3 Model Comparison - R² Score graph

In contrast, XGBoost shows the lowest performance, with an R² score of just 30.76% and MAE of 0.32. Ridge Regression improves upon this with 46.79% R² and a slightly better MAE of 0.26. Random Forest Regressor performs notably well among the individual models with a 76.12% R² score and 0.21 MAE, but still falls short of the stacking model. These findings demonstrate that ensemble techniques, particularly stacking, can effectively combine the strengths of multiple models to deliver superior performance. Therefore, the stacking model is a reliable and practical approach for real-world agricultural price forecasting, providing both high accuracy and consistency. Following Fig.4 shows analysis of actual vs predicted price.

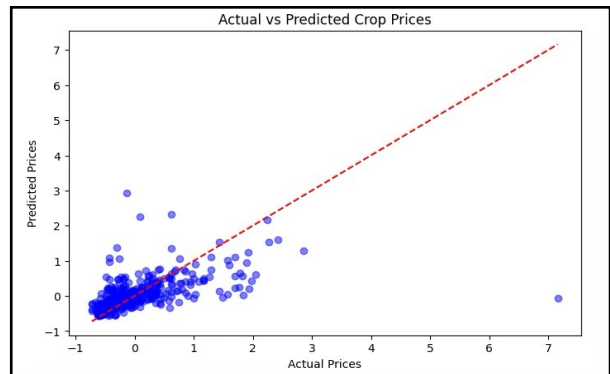


Fig.4 Actual vs Predicted Price

The scatter plot, in which the red dashed diagonal line ($y = x$) denotes perfect forecasts and each blue dot represents a data point, shows the link between actual and projected crop prices. The majority of the data points are tightly packed around this line, indicating that the model does a good job of capturing general pricing trends. However, some outliers show some variation in prediction accuracy for extreme scenarios, especially at higher actual price values. Despite this, the model's great predictive capacity is confirmed by the overall alignment of points along the diagonal, which supports the results' low MAE and high R² score.

The research proved that a Stacking Ensemble Regression model where XGBoost acted as the meta-learner and Random Forest and Ridge Regression were used as the base learners performed much better than conventional single-model strategies, with an R^2 of 91.36% and MAE of 0.18. Although XGBoost was not great as a standalone model most probably because it is noise-sensitive in the data, sensitive to minor fluctuations, and susceptible to selecting for high-order nonlinear interactions over linear ones it performed extremely well in an ensemble context. As a meta-learned model, XGBoost utilized Random Forest's heterogenous prediction capabilities (nonlinear feature identification) and Ridge Regression's linear dependency modeling to pool their complementary predictions to generate highly accurate and stable predictions. This also further identifies the novel contribution of the proposed method: the conversion of individual model weaknesses to collective strengths, thereby offering a scalable, reliable, and viable methodology for crop price forecasting which can guide more informed decisions by farmers, policymakers, and market players.

REFERENCES

1. Dutt, Sandhu, et al. "Agricultural price prediction through artificial intelligence." *International Journal of Creative Research Thoughts (IJCRT)* 12.2 (2024). <https://doi.org/10.37118/ijdr.28018.03.2024>
2. Rana, Humaira, et al. "Prediction of agricultural commodity prices using big data framework." *Engineering, Technology & Applied Science Research* 14.1 (2024): 12652-12658. <https://doi.org/10.48084/etasr.6468>
3. Sharma, Chandan, et al. "Price prediction model of fruits, vegetables and pulses according to weather." 2023 13th international conference on cloud computing, data science & engineering (confluence). IEEE, 2023. <https://doi.org/10.1109/Confluence56041.2023.10048880>
4. Thapaswini, G., and M. Gunasekaran. "A methodology for crop price prediction using machine learning." 2022 IEEE 2nd international conference on mobile networks and wireless communications (icmnwc). IEEE, 2022. <https://doi.org/10.1109/ICMNWC56175.2022.10031852>
5. Dhanapal, Ranjani, et al. "Crop price prediction using supervised machine learning algorithms." *Journal of physics: Conference series*. Vol. 1916. No. 1. IOP Publishing, 2021. <https://doi.org/10.1088/1742-6596/1916/1/012282>
6. Zheng, Guang, et al. "The research on agricultural product price forecasting service based on combination model." 2020 IEEE 13th international conference on cloud computing (cloud). IEEE, 2020. <https://doi.org/10.1109/CLOUD49709.2020.00009>
7. Jain, Ayush, et al. "A framework for crop price forecasting in emerging economies by analyzing the quality of time-series data." *arXiv preprint arXiv:2009.04171* (2020). <https://doi.org/10.48550/arXiv.2009.04171>
8. Hegde, Girish, Vishwanath R. Hulipalled, and J. B. Simha. "Price prediction of agriculture commodities using machine learning and NLP." 2021 second international conference on smart technologies in computing, electrical and electronics (ICSTCEE). IEEE, 2021. <https://doi.org/10.1109/ICSTCEE54422.2021.9708582>
9. Bali, Nishu and Anshu Singla. "Deep Learning Based Wheat Crop Yield Prediction Model in Punjab Region of North India." *Applied Artificial Intelligence* 35 (2021): 1304 - 1328. <https://doi.org/10.1080/08839514.2021.1976091>
10. Shahhosseini, Mohsen et al. "Forecasting Corn Yield With Machine Learning Ensembles." *Frontiers in Plant Science* 11 (2020): n. Pag. <https://doi.org/10.3389/fpls.2020.01120>
11. Zhang, Dabin et al. "Forecasting Agricultural Commodity Prices Using Model Selection Framework With Time Series Features and Forecast Horizons." *IEEE Access* 8 (2020): 28197-28209. <https://doi.org/10.1109/ACCESS.2020.2971591>
12. Elavarasan, Dhivya et al. "Crop Yield Prediction Using Deep Reinforcement Learning Model for Sustainable Agrarian Applications." *IEEE Access* 8 (2020): 86886-86901. <https://doi.org/10.1109/ACCESS.2020.2992480>
13. Mishra, Akshansh. "Machine Learning Classification Models for Detection of the Fracture Location in Dissimilar Friction Stir Welded Joint." *Applied Engineering Letters : Journal of Engineering and Applied Sciences* (2020): n. pag. <https://doi.org/10.18485/aeletters.2020.5.3.3>
14. Haider, Sajjad Ali et al. "LSTM Neural Network Based Forecasting Model for Wheat Production in Pakistan." *Agronomy* (2019): n. Pag. <https://doi.org/10.3390/AGRONOMY9020072>
15. Young, Linda J.. "Agricultural Crop Forecasting for Large Geographical Areas." *Annual Review of Statistics and Its Application* (2019): n. Pag. <https://doi.org/10.1146/ANNUREV-STATISTICS-030718-105002>
16. Basso, Bruno, and Lin Liu. "Seasonal crop yield forecast: Methods, applications, and accuracies." *Advances in agronomy* 154 (2019): 201-255. <https://doi.org/10.1016/bs.agron.2018.11.002>
17. Kaggle, "Crop Prediction Dataset" Kaggle Datasets, 2022. [Online] Available: <https://www.kaggle.com/datasets/gokulprasanht/crop-prediction>